

On the Feasibility of EOG-Based Eye-Writing in Virtual Reality

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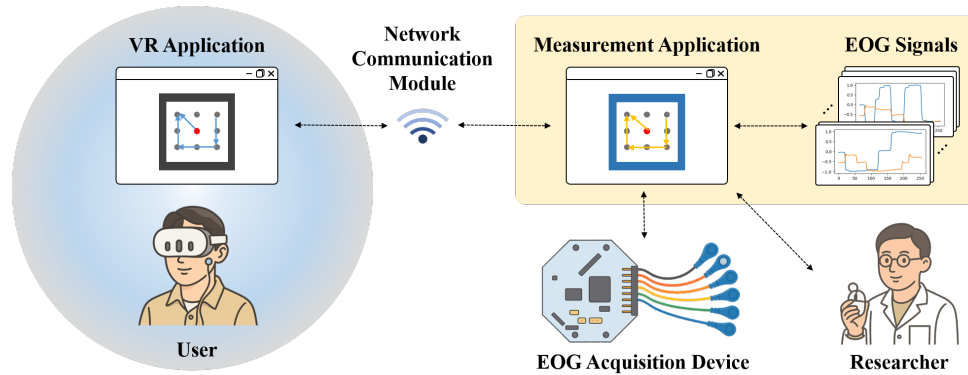


Figure 1: System architecture of the proposed VR-based EOG eye-writing interface.

ABSTRACT

Virtual reality (VR) offers immersive environments for exploring gaze-based input modalities beyond conventional hand- or controller-based interaction. This work investigates the feasibility of electrooculography (EOG)-based eye-writing as a gaze-based interface in VR, addressing limitations of prior monitor-based approaches that restrict natural movement. We developed a VR-guided acquisition system integrating a customizable VR interface with a desktop-based measurement application for synchronized EOG data collection. Using this system, we collected EOG data from 18 participants performing digit eye-writing tasks (0–9). Our results show that VR-based acquisition can capture meaningful EOG patterns corresponding to eye-writing gestures.

Index Terms: Virtual Reality (VR), Human-Computer Interaction (HCI), Electrooculography (EOG), Eye-writing interfaces, Assistive communication systems.

1 INTRODUCTION

Virtual reality (VR) provides immersive environments that preserve users' natural head and eye movements, making it an attractive platform for gaze-based human-computer interaction (HCI) and assistive communication [3].

Electrooculography (EOG)-based eye-writing enables hands-free character input using eye movements and has been validated primarily in monitor-based setups, where head movement is restricted [2]. Such constraints limit ecological validity and may not reflect natural gaze behavior [1]. EOG provides a lightweight alternative to camera-based eye tracking and is less affected by illumination changes and pupil visibility issues.

In this work, we investigate the feasibility of EOG-based eye-writing in VR. We developed a VR-guided acquisition system

and collected EOG data from 18 participants performing digit eye-writing tasks (0–9). Using this dataset, we report initial recognition results from eye-writing experiments conducted in VR.

2 MATERIALS AND METHODS

2.1 VR and EOG Measurement Applications

The proposed system consists of a VR application and a desktop-based EOG measurement application. The VR application, running on a Meta Quest 3 headset, presents visual patterns for eye-writing tasks, while the measurement application remotely controls the VR interface and synchronizes EOG data acquisition processes. EOG signals are recorded using an OpenBCI Cyton board, as illustrated in Fig. 1.

The VR interface supports light and dark modes and multilingual guidance with a voice toggling option to improve user comfort. Before data collection, participants completed a brief practice phase with visual cues, followed by a test phase without cues to ensure consistent eye-writing behavior (Fig. 2).

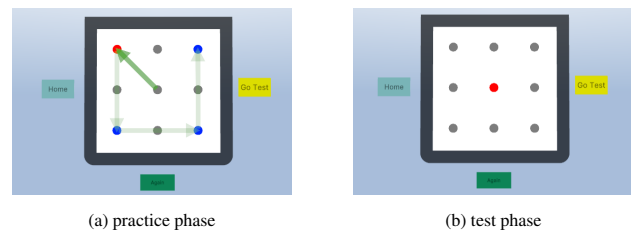


Figure 2: VR interface for eye-writing tasks.

2.2 Data Collection

Data from 18 participants (8 males and 10 females; aged 20–38 years) were used in this study. EOG signals were acquired using an OpenBCI Cyton board with six electrodes placed around the eyes and on the mastoids (Fig. 3), while participants wore a VR headset for visual guidance.

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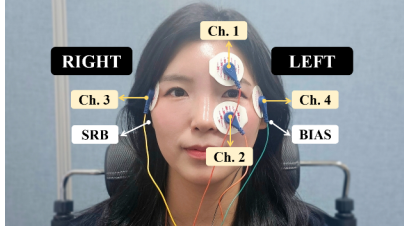


Figure 3: Electrode placement for EOG acquisition.

Participants performed eye-writing tasks by tracing ten numerical patterns (digits 0-9), each repeated three times, within the VR interface (Fig. 4). Before each trial, participants fixated on a central reference point before following the predefined trajectory using eye movements. Each trial included a 5-second eye-writing period with rest intervals. Signals were recorded at 250 Hz, and the study protocol was approved by the IRB.

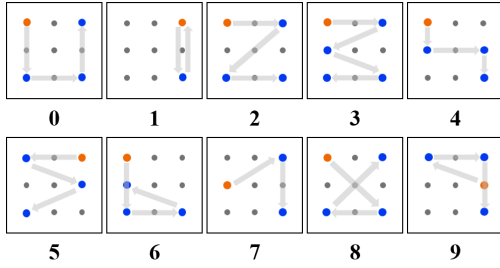


Figure 4: Eye-writing patterns used for EOG data collection.

2.3 EOG Processing and Recognition

Raw EOG signals (0–50 Hz) were preprocessed to reduce noise and inter-trial variability. The signals were downsampled (250 Hz to 64 Hz) and median-filtered to suppress high-frequency noise. Eye-blink artifacts were detected on the vertical channel using amplitude thresholding and mitigated via interpolation, and baseline drift was corrected using the initial fixation period. Horizontal and vertical components were then derived from differential electrode pairs, and each trial was normalized to a fixed range before analysis.

The preprocessed EOG sequences were used as input to a convolutional recurrent neural network with an attention mechanism (CRNN-Attention). Recognition performance was examined using a subject-independent train–test split.

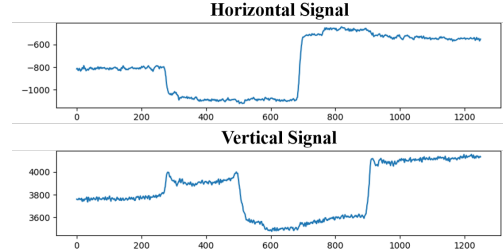
3 RESULTS

3.1 Visualization of Eye-Writing Patterns

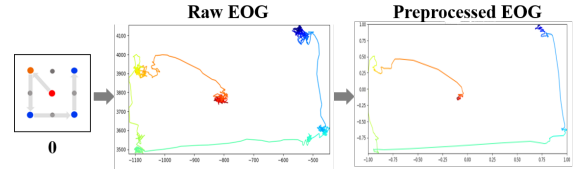
To qualitatively validate the VR-based EOG dataset, we visualized representative EOG signals and reconstructed a 2D trajectory for the digit “0” (Fig. 5), showing patterns consistent with the intended eye-writing motion. This visualization suggests that the VR-guided interface elicited clear EOG responses during eye-writing.

3.2 Classification Results

Using a subject-independent train–test split, we evaluated the proposed classification pipeline on the collected VR-based EOG dataset. We fixed the subject split by randomly selecting 15 participants for training and 3 participants for testing (seed = 42). The model was trained and evaluated 10 times with different initializations, and the averaged results are reported. The model achieved an average accuracy of $96.22\% \pm 2.88\%$



(a) Raw horizontal/vertical signals



(b) Reconstructed 2D EOG trajectory from raw and preprocessed signals

Figure 5: Visualization of EOG signals for the digit “0”.

and a macro-F1 score of $96.10\% \pm 3.06\%$, indicating consistent recognition performance across different random splits.

4 DISCUSSION AND CONCLUSION

This work examined the feasibility of EOG-based eye-writing in virtual reality environments. We developed a VR-guided acquisition system and collected EOG data from 18 participants performing digit eye-writing tasks. Qualitative visualizations and recognition experiments indicate that VR-based acquisition can capture meaningful eye-writing patterns suitable for further analysis.

Compared to conventional monitor-based setups, VR enables more natural gaze behavior and immersive guidance, which may improve user comfort and ecological validity. While natural head movement introduces additional variability, the results suggest that EOG-based eye-writing remains feasible within VR environments.

Despite limitations such as a modest sample size and sensitivity to motion-related noise, this study demonstrates the potential of VR-based EOG eye-writing as a gaze-based input modality. Future work will focus on expanding the dataset, improving robustness through signal processing and compensation techniques, including additional baselines and advanced models, and further exploring user-centered interface designs to enhance comfort and accessibility.

ACKNOWLEDGMENTS

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